THE UNIVERSE ON THREE WHALES

How Ancient Myths Predicted Modern Cosmology

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Contents

Introduction: Why Myths Are Always Right **1**

1 Three Whales Holding the World **2**
1.1 Ancient Wisdom . . . 2
1.2 Modern Understanding . . . 2
1.3 Why This Matters . . . 3

2 First Whale—Form (The Donut Universe) **4**
2.1 What the Ancients Saw . . . 4
2.2 What Science Sees . . . 4
2.3 How to Test It? . . . 5
2.4 What Does This Give Us? . . . 5

3 Second Whale—Motion (The Great Drain) **6**
3.1 Myth of the Eternal Cycle . . . 6
3.2 Scientific Discovery . . . 6
3.3 Mathematics of the Miracle . . . 7
3.4 Why Don't We See the "Drain"? . . . 7
3.5 Solving Mystery #2 . . . 7

4 Third Whale—Structure (The Cosmic Sponge) **8**
4.1 Myth of the Sponge and Honeycombs . . . 8
4.2 What the Telescope Shows . . . 8
4.3 The Brilliant Discovery . . . 8
4.4 Mathematics of the Sponge . . . 9
4.5 Why This Matters . . . 9
4.6 Solving Mystery #3 . . . 9

5 The Kobrin Trace: From Zariski to IT3 **10**
5.1 Personal History . . . 10
5.2 Synchronicity . . . 10
5.3 What Does This Mean? . . . 11

6 What This Means for Each of Us **12**
6.1 Philosophical Meaning 12
6.2 Practical Implications 12
6.3 What's Next? 12

Conclusion: The Circle is Closed **14**

Introduction: Why Myths Are Always Right

Ancient people said: "The Earth rests on three whales."

Modern scientists laughed: "It's a naive myth!"

But what if they were both right?

What if the ancients saw not the physical Earth, but the **structure of Reality itself**? What if their "three whales" were a coded description of the three fundamental principles holding the Universe together?

This book is not just about cosmology. It's about how the **intuition of humanity**, expressed in myths, met the **mathematical rigor** of modern science. And they recognized each other.

I will show you that the Universe truly stands on three "whales". But they are not animals; they are **three pillars of existence**:

1. **Form**—The Universe is finite and shaped like a donut (a torus);
2. **Motion**—Energy constantly "drains" toward the topological center;
3. **Structure**—Matter is distributed like a sponge with voids.

And the most amazing part: this model isn't just beautiful. It **solves the major mysteries of modern cosmology** without inventing "dark energy" or other entities.

Get ready for a journey where mathematics meets mythology, and cosmology meets destiny.

Chapter 1

Three Whales Holding the World

1.1 Ancient Wisdom

In Slavic mythology, there's a beautiful image: the Earth rests on three enormous whales swimming in a primordial ocean. When these whales grow tired and shift, earthquakes occur.

Seems like naive ancient fantasy. But let's look closer.

Three whales. Not two, not four. Exactly three.

Why? Because three is the **minimum number that creates stability**. Two is unstable, four is excessive. Three is the golden mean, a principle of triality that permeates the Universe:

- Space has three dimensions;
- Time divides into past, present, and future;
- Matter consists of three quarks;
- Colors—from three primaries (red, green, blue).

The ancients intuitively felt: **stability requires triality**.

1.2 Modern Understanding

Now let's leap to the 21st century. I studied the cosmic microwave background radiation—the "echo" of the Big Bang. And I discovered something remarkable: the Universe truly stands on **three fundamental principles**:

First Whale—TOPOLOGY (Form)

The Universe is not infinite. It is finite and closed, like the surface of a donut. If you fly straight, you'll return to where you started. This explains why there's "missing" energy on large angular scales in the cosmos—the waves simply don't fit entirely inside our "donut".

Second Whale—THERMODYNAMICS (Motion)
The Universe is not static. There is a constant "drain" of energy: when galaxies collide, energy is released, but instead of dissipating into infinity, it **focuses toward the center** of the fundamental domain. This flow creates negative pressure that pushes space apart—what we mistakenly call "dark energy".

Third Whale—STRUCTURE (Sponge)
The Universe is not homogeneous. It resembles a sponge: 75 % of the volume is voids, while matter is gathered into thin filaments and nodes. This "sponginess" is not accidental—it acts as a system of channels through which energy flows efficiently to the center, amplifying the effect by a factor of $\xi \approx 3.8$!

Three whales. Just as the ancients said.

1.3 Why This Matters

When we realize that myths are not fabrications but **encoded truths**, everything changes. The ancients were not less intelligent than us. They simply spoke a different language—the language of images and metaphors.

Their "three whales" are not zoology. It is **cosmology**.

And now that we have mathematical equations confirming this triality, we can say: **the myths were right**. It just took science two thousand years to catch up with intuition.

Chapter 2

First Whale—Form (The Donut Universe)

2.1 What the Ancients Saw

Many cultures feature a symbol of a **circle closing upon itself**:

- The Ouroboros—a serpent eating its own tail;
- Celtic knots without beginning or end;
- Mandalas with a central point;
- Indian yantras with nested circles.

These are not mere decorations. They are an **intuitive understanding of a finite, closed Universe**.

2.2 What Science Sees

In 2026, I published a paper showing that data from the Planck space telescope perfectly matches the **Flat Irrational Torus (IT3) model**.

What does this mean in plain words?

Imagine a video game where a character exiting the right edge of the screen appears on the left. Exiting the top appears at the bottom. This is **toroidal topology**—space closed upon itself.

Our Universe works the same way! Only in three dimensions.

Size of the "donut": approximately 28.6 billion light-years.

Why "irrational"? Because the side ratios are not simple numbers, but $\sqrt{2}$ and $\sqrt{3}$. This makes the structure stable and prevents pattern overlap—like in music, where irrational frequencies create rich, non-monotonous sound.

2.3 How to Test It?

If the Universe is finite, light from distant objects can reach us via **multiple paths**—as if you were looking in a mirror and seeing multiple reflections of yourself.

In the CMB, this manifests as **"matched circles"**—regions with identical temperatures arranged symmetrically. For our model, the predicted radius of such circles is 30°–60°.

This can be tested! And the next generation of telescopes (CMB-S4, Euclid) will do just that.

2.4 What Does This Give Us?

Solving Mystery #1: Why is there less energy on large angular scales (low multipoles) than theory predicts?

Answer: Because the waves simply **don't fit** entirely inside our "donut"! There's a minimum wavelength determined by the size of the Universe. Anything longer gets "cut off".

It's like trying to play a note lower than your flute allows. It won't work!

First whale found. Form determined.

Chapter 3

Second Whale—Motion (The Great Drain)

3.1 Myth of the Eternal Cycle

All mythologies contain the idea of **cyclicality**:

- Indian kalpas (cycles of creation and destruction);
- The Greek wheel of time;
- The Slavic Kolo Svaroga;
- The Daoist principle "everything flows, everything changes".

The ancients felt: The Universe is not static. There is **flow**, motion, energy exchange.

3.2 Scientific Discovery

When I studied the data, a problem arose: if the Universe is finite, why doesn't it collapse back under gravity? And why doesn't it overheat from energy released during galaxy mergers?

The answer turned out to be brilliantly simple: **energy doesn't accumulate—it "drains"**.

Imagine a sink. Water flows from the tap (galaxy mergers), swirls down the funnel (topological structure), and **goes into the drain** (center of the fundamental domain).

This "drain" creates **negative pressure**—like pulling a plug from a bathtub. Water starts moving not because it's being pushed, but because **it's being "sucked out"**.

3.3 Mathematics of the Miracle

I derived equations describing this process:

Sink pressure: $P = -\kappa \times \rho^2$,

where κ depends on the size of the Universe, and ρ—matter density.

Dissipation rate: $\Gamma = \eta \times (G \times L)/c$,

where G—gravitational constant, L—topology size, c—speed of light.

These equations show: the more compact the Universe, the more efficient the "drain". And this process **naturally accelerates expansion** without any "dark energy"!

3.4 Why Don't We See the "Drain"?

Because it's **topological**, not local. It's not a black hole you can fall into. It's a **geometric effect** of the entire space-time structure.

Like an ant on a donut's surface doesn't see the donut's hole, but feels its influence on geometry.

3.5 Solving Mystery #2

Mystery: What is "dark energy" pushing galaxies apart?

Answer: There is no dark energy! There is a **topological energy sink** creating effective negative pressure.

It's like thinking someone is pushing water in a sink, when actually it's being **drained**.

Second whale found. Motion explained.

Chapter 4

Third Whale—Structure (The Cosmic Sponge)

4.1 Myth of the Sponge and Honeycombs

Recall myths about the **world tree** connecting different levels of reality. Or the **web of creation**, where every thread is linked to others.

This isn't just poetry. It's a description of the **structure of matter**.

4.2 What the Telescope Shows

When astronomers mapped galaxy distribution, they discovered something amazing: the Universe resembles a **sponge** or **Swiss cheese**.

75 % **of the volume**—voids hundreds of millions of light-years across.
25 %—thin filaments of galaxies connecting into nodes (clusters).

This isn't chaos. It's an **ordered porous structure** with fractal dimension ~ 2.6.

4.3 The Brilliant Discovery

And here's the miracle. I realized: this "sponginess" isn't a side effect. It's a **functional necessity**!

Imagine needing to deliver energy from one point of the donut to another. If space were uniformly filled with matter, energy would scatter, collide, and be lost.

But with **channels** (voids), energy flows through them like water in pipes, virtually losslessly!

4.4 Mathematics of the Sponge

I introduced an **amplification coefficient** ξ (xi), showing how many times the sponge structure amplifies energy flow:

$$\xi = \left(\frac{\phi}{1-\phi}\right)^{\alpha}$$

where ϕ—porosity (void fraction), $\alpha \approx 1.2$.

For our Universe ($\phi = 0.75$): $\xi \approx 3.8$.

This means the sponge structure **amplifies the sink effect nearly 4 times**! Without it, the mechanism wouldn't be efficient enough.

4.5 Why This Matters

This explains **why the sink is active now but wasn't in the early Universe**:

- Early Universe: homogeneous, $\phi \approx 0 \Rightarrow \xi \approx 1$ (weak sink);
- Modern Universe: sponge-like, $\phi \approx 0.75 \Rightarrow \xi \approx 3.8$ (strong sink).

Nature itself "turned on" accelerated expansion once large-scale structure formed!

4.6 Solving Mystery #3

Mystery: Why did accelerated expansion start only a few billion years ago?

Answer: Because it took time for the **sponge structure** to form with sufficient porosity. Before that, the sink was too weak.

Third whale found. Structure understood.

Chapter 5

The Kobrin Trace: From Zariski to IT3

5.1 Personal History

This book would be incomplete without the story of how **destiny intertwines with science**.

I was born in Kobrin, Belarus. An ordinary town few have heard of. But one day I learned an amazing fact: in this same town in 1899 was born **Oscar Zariski**—one of the greatest mathematicians of the 20th century, founder of modern algebraic geometry and topology.

Zariski left Kobrin young, revolutionized mathematics, but his name is nearly forgotten at home. I decided to fix this: studied his biography, gathered materials, working to install a memorial plaque in Kobrin.

5.2 Synchronicity

And here the magic of numbers begins.

I received a letter from Professor Jean-Pierre Luminet—a pioneer of compact-space cosmology. And I discovered: **we were born on the same day—June 3!**

Luminet: June 3, 1951

Me: June 3, 1988

Exactly 37 years apart.

When I connected these dots:

- **Zariski (1899, Kobrin)**—created mathematical topology;
- **Luminet (1951, June 3)**—applied topology to cosmology;
- **Me (1988, June 3, Kobrin)**—unified their ideas in the IT3 model.

...I understood: this is no accident. It's a **topological loop**, closed across time and space.

5.3 What Does This Mean?

The ancients said: "Everything is connected to everything".
Modern physics says: "Everything correlates with everything".
Topology says: "Everything is connected by invisible threads".

My work—this isn't just science. It's **repaying a debt** to the place where I was born, and to the people who paved the way.

Zariski left Kobrin to give the world topology.
I returned to Kobrin to return Zariski to it.

The circle is closed.

Chapter 6

What This Means for Each of Us

6.1 Philosophical Meaning

When we understand that the Universe is:

- **Finite** (not infinite and frightening);
- **Stable** (won't collapse or overheat);
- **Structured** (not chaos, but an ordered sponge).

...our perception of ourselves within it changes.

6.2 Practical Implications

1. We are not alone in infinity

The Universe is finite. This means everything that exists is **interconnected**. There's no "somewhere far away" in an absolute sense. We are part of a single whole.

2. Energy is not lost

The sink mechanism shows: the Universe is a **closed system with efficient energy recycling**. Nothing is wasted.

3. Structure matters

Sponge architecture isn't accidental; it's a **functional necessity**. So too in life: what matters isn't just the "nodes" (events), but the "voids" (pauses between them).

6.3 What's Next?

Scientific path:

- 2027–2030: Telescopes CMB-S4 and Euclid will search for "matched circles";

- If found—the IT3 model is confirmed;
- If not—we must look for another topology.

Personal path:

- Continue work on restoring Zariski's name;
- Develop the "cosmic sponge" idea in other fields (neurobiology, sociology);
- Seek new synchronicities.

Human path:

- Remember: we are part of something greater;
- Value structure and void equally;
- Believe that myths and science are two languages of one truth.

Conclusion: The Circle is Closed

The ancients said: "The Earth rests on three whales."

We said: "The Universe stands on three pillars: Form, Motion, Structure."

It's the same thing. Just different words.

Three whales found:

1. **Topology**—The donut Universe;
2. **Energy sink**—The Great Drain;
3. **Sponge structure**—The cosmic web.

Three mysteries solved:

1. Why is energy missing on large scales? → Topological infrared cutoff;
2. What is dark energy? → Sink effect amplified by the sponge;
3. Why did acceleration start recently? → Structure finally formed.

Three lives connected:

1. Oscar Zariski (1899, Kobrin);
2. Jean-Pierre Luminet (1951, June 3);
3. Victor Logvinovich (1988, June 3, Kobrin).

Myths were right. Science confirmed it.

Now it's your turn to decide: believe in an infinite, cold, meaningless Universe... or see a **finite, stable, meaningful world**, standing on three whales of topology.

The choice is yours.

But remember: **everything is connected to everything**. Even if you don't see it.

Kobrin, 2026